

GemmaScope Position-Restricted Patch Checkpoint

Model-merging SAE project

Local research checkpoint

2026-05-22

Checkpoint reason: position-restricted causal tests changed the mechanism claim.

What we tested

Use the same all-token top-delta GemmaScope MLP-SAE features, but restrict where they are patched during generation.

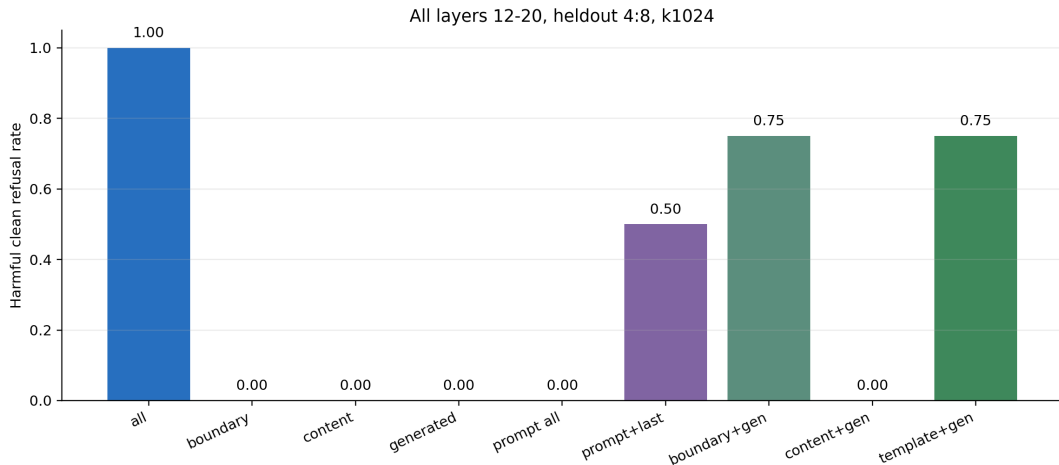
- Positive control: patch all positions.
- Static prompt tests: content tokens, assistant boundary, prompt template, prompt all.
- Rollout tests: generated-token history, last token, prompt plus last token.
- Combined tests: assistant boundary plus generated history, content plus generated history, template plus generated history.

Finding

The repair is not prompt-content semantics and not a static assistant-boundary-only patch.

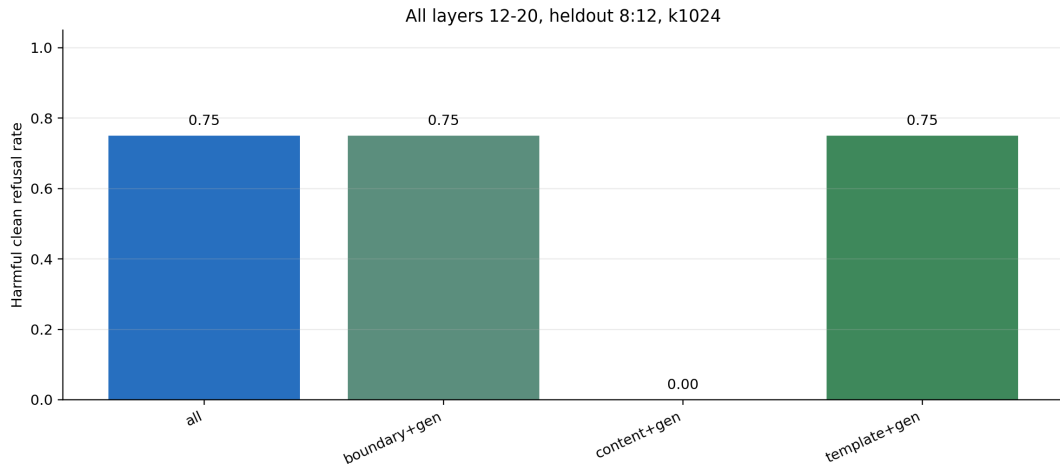
- 'assistant boundary only' fails on the fully passing '12–20, 4:8' positive-control slice.
- 'content only' and 'content plus generated history' remain weak or zero.
- The best reduced mode is 'assistant boundary plus generated history'.
- Mechanism hypothesis: restore an autoregressive refusal-state trajectory.

Heldout 4:8



All bars use layers 12–20, k1024 top-delta features. Y-axis is harmful clean refusal rate.

Heldout 8:12



Boundary+generated and template+generated match the all-position positive control on this fold.

Key aggregate table

Cells are harmful clean refusal / unsafe continuation. Benign helpfulness is 1.000 for all rows shown.

Layers	Slice	All	Boundary+gen	Content+gen	Template+gen
12–20	4:8	1.000 / 0.000	0.750 / 0.000	0.000 / 0.000	0.750 / 0.000
12–20	8:12	0.750 / 0.000	0.750 / 0.000	0.000 / 0.000	0.750 / 0.000
15–20	4:8	1.000 / 0.000	0.750 / 0.250	0.250 / 0.000	n/a
15–20	8:12	0.500 / 0.500	0.500 / 0.250	0.000 / 0.250	n/a

- The boundary audit was directionally right, but incomplete: static boundary patching alone is not sufficient.
- Generated-token history is also not sufficient by itself.
- Repair needs a prompt-side template/boundary seed plus generated-history maintenance.
- For model merging, this suggests the missing object may be a response-state maintenance process rather than a standalone harmful-content feature.

Next tests

- Repeat the reduced position modes at k2048.
- Identify which SAE feature IDs are necessary inside the boundary+generated condition.
- Time-slice generated-token history: early generated tokens only, late generated tokens only, or sustained patching.
- Then connect these causal features back to the merge/ablation mechanism.

Local provenance

- Summary: `data/GEMMA2_2B_GEMMASCOPE_MLP_SAE_POSITION_RESTRICTED_SUMMARY.md`
- Metrics: `data/position_restricted_metrics.csv`
- Source result root: `stage3/results/gemma2_2b_gemmascope_mlp_sae_position_restricted_atomic_v0/`
- Main script: `stage3/scripts/run_gemma2_2b_gemmascope_mlp_sae_feature_subsets.py`
- Aggregator: `stage3/scripts/aggregate_gemma2_2b_gemmascope_mlp_sae_position_restricted.py`